

Mohtashim Butt

mohtashim.lums@gmail.com | +92 306 4588419 | [LinkedIn](#) | [GitHub](#)

EDUCATION

Lahore University of Management Sciences – BS Computer Sciences

Lahore, Pakistan

Relevant Courses: Computer Vision, Machine Learning, Language Processing with Generative AI, Deep Learning, Introduction to AI, Software Engineering, Data Structures & Algorithms

Aug. 2020 – May 2024

PUBLICATIONS

Geographically Generalisable Urban Green Space Segmentation for Developing Countries

Aug. 2025

Mohtashim Butt, Nawaal Siddique, Fatima Naweed, Zubair Khalid

Conference: IEEE International Geoscience and Remote Sensing Symposium, Brisbane, Australia | [Paper](#)

PROFESSIONAL EXPERIENCE

AI & Automation Engineer

Acorn IT Services LLC.

California, USA (Remote)

Nov. 2025 – Present

- Built a full-stack data platform end-to-end: API layer, a 21-table RDS PostgreSQL data model, and an AWS Bedrock/Claude inference stage, turning raw operational data into automated per-client analytical reports.
- Designed the data ingestion/pre-processing (PII-scrubbing) pipeline feeding it, engineering it to run at **\$50/mo** against a \$250 budget.
- Shipped **10+** backend automation services integrating multiple REST APIs (Autotask, Datto RMM) across a **70-client** production environment, cutting **~350 hrs/month** of manual work.
- Engineered an end-to-end monitoring pipeline (API polling → FastAPI → InfluxDB → Grafana) for real-time job health across **70** clients, saving \$100/mo.
- Led experiment design/evaluation for an AI ticket-classification model across **8** queues and **257** skills, using data analysis to reach **~90%** routing accuracy while keeping compliance-sensitive paths deterministic.
- Diagnosed and resolved L1–L3 infrastructure issues across **70+** client environments, using recurring failure data to prioritize automation targets.

Software Engineer 1

Careem (Uber Inc.)

Dubai, UAE (remote)

Aug. 2024 – Sept. 2025

- Owned full-stack quality on a production backend microservice: drove test coverage to **90%**, managed SLO/SLA latencies, mitigated security vulnerabilities, and built out Dynatrace log/metrics monitoring.
- Rebuilt an internal tool's front-end on a TypeScript + Vite micro-frontend architecture, improving scalability and module-level ownership.
- Designed and maintained a centralized data/health dashboard spanning several backend microservices, combining SQL-based data analysis with a Python and internal-tooling backend.

Machine Learning Intern

Centre for Water Informatics and Technology

Lahore, Pakistan

May. 2023 – June. 2023

- Built a full-stack embedded system: four ESP-32 cameras feeding a self-hosted backend (PHP/Python/Arduino), and trained/fine-tuned a tiny-YOLOv5 model on the captured frames for real-time forest-fire detection. | [Report](#)

PROJECTS

- Cloud-Native Ride-Hailing Microservices:** Designed and deployed 5 containerized backend services on AWS ECS Fargate behind an internal ALB with ECR and path-based API routing, plus an S3 + CloudFront frontend | [Code](#)
- Automatic Annotation Tool via Semi-Supervised Learning:** Trained/fine-tuned a DeepLabv3-ResNet model on manual petrograph annotations to auto-generate masks and label JSONs for unseen images from a single bounding box. | [Code](#)
- Lane Segmentation for Autonomous Driving:** Combined a LaneNet segmentation model with YOLOv7 vehicle detection and a dynamic homography step to render a real-time orthographic top-view of the lane. | [Code](#)

TEACHING EXPERIENCE

- Deep Learning (CS-5312) — Teaching Assistant, Spring-2024:**
- Computer Vision Fundamentals (CS-5310) — Teaching Assistant, Fall-2023:**

LANGUAGES & TECHNOLOGIES

- Languages & Data:** Python, C/C++, Go, TypeScript, SQL, Bash, PHP; PostgreSQL, MongoDB, InfluxDB.
- Cloud, ML & Tooling:** AWS (Lambda, Bedrock, RDS, S3, ECS, IAM, SES), Docker, Git, FastAPI, Grafana, PyTorch, scikit-learn, Hugging Face, OpenCV, REWST, n8n, Autotask PSA, Datto RMM.